

TRIP DOWN MEMORY LANE

411410065 鄭俊良

412410001 陳達翔

412410003 劉靖媛

THE PROBLEM

- lonely and disconnected from family
- memory decline (dementia)



OUR SOLUTION

- **WHAT**
 - system for preserving memories
- **HOW**
 - uses AI for image analysis, story prompting & quiz generation
- **WHY**
 - achieves cognitive stimulation and family connection



PROJECT SCOPE

- **Target**
 - elders and their families
- **Focus**
 - photo-based memories (3-photo requirement)
 - 3-question multiple choice quiz only
- **Data Storage**
 - data handling client-side (IndexedDB)



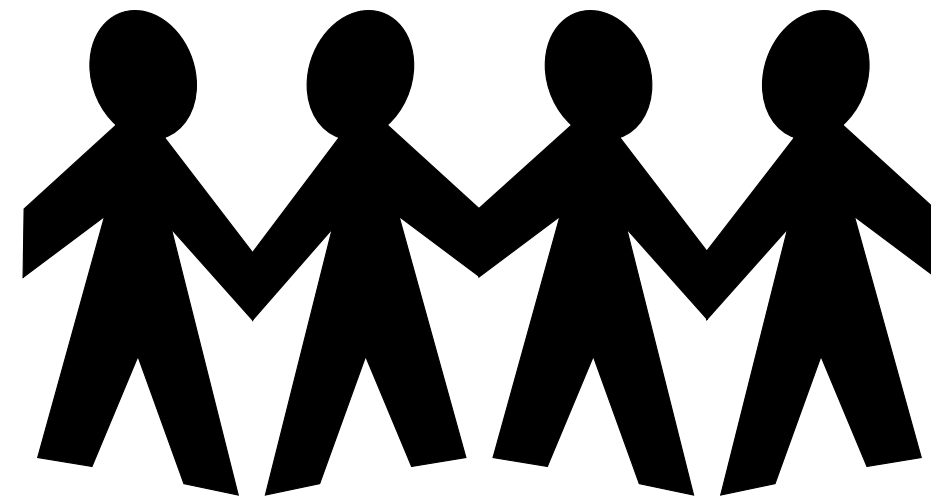
KEY FEATURES

- Upload picture & edit memory story
- AI-generated interactive quiz
- Self record or Text-to-speech(TTS) for accessibility
- Elder-friendly UI/UX
 - interactive card design
 - big font size
 - simple buttons



APPLICATIONS

- home use for family bonding
- elder care facilities
- cognitive training programs



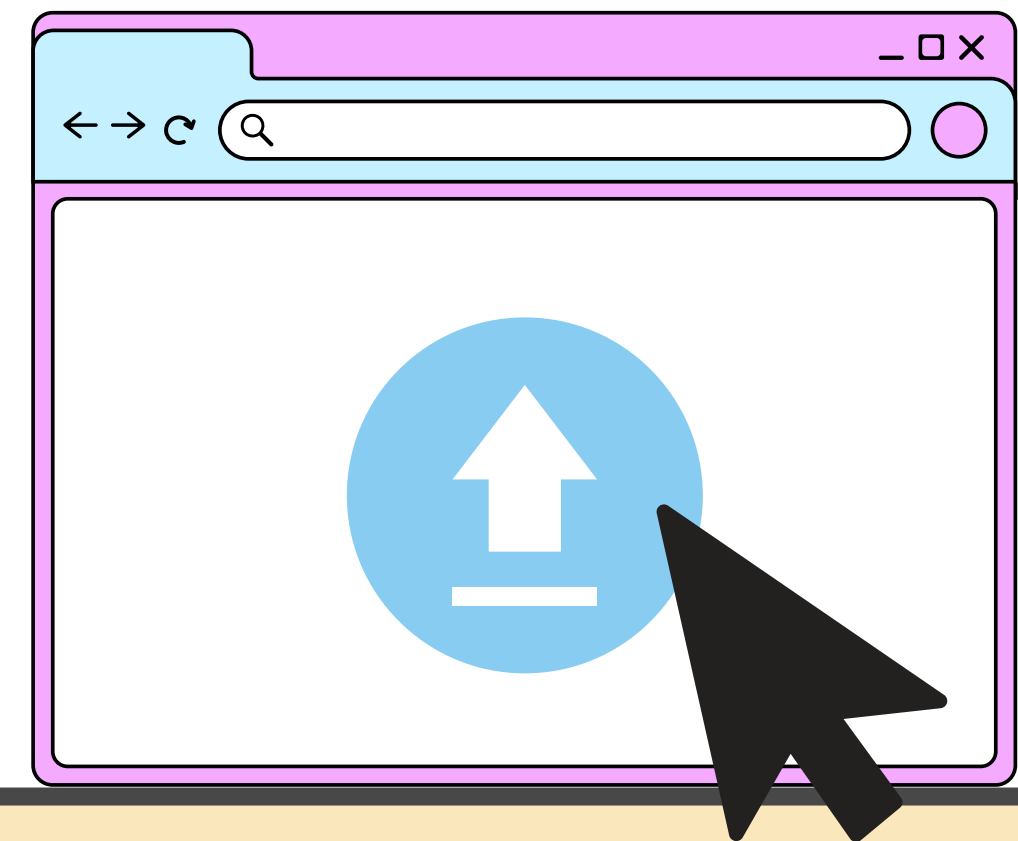
CURRENT MODULES USED

- **Google Vision API**
 - image analysis
- **Hugging Face Inference API**
 - text generation
- **Google Cloud TTS**
 - voice audio generation

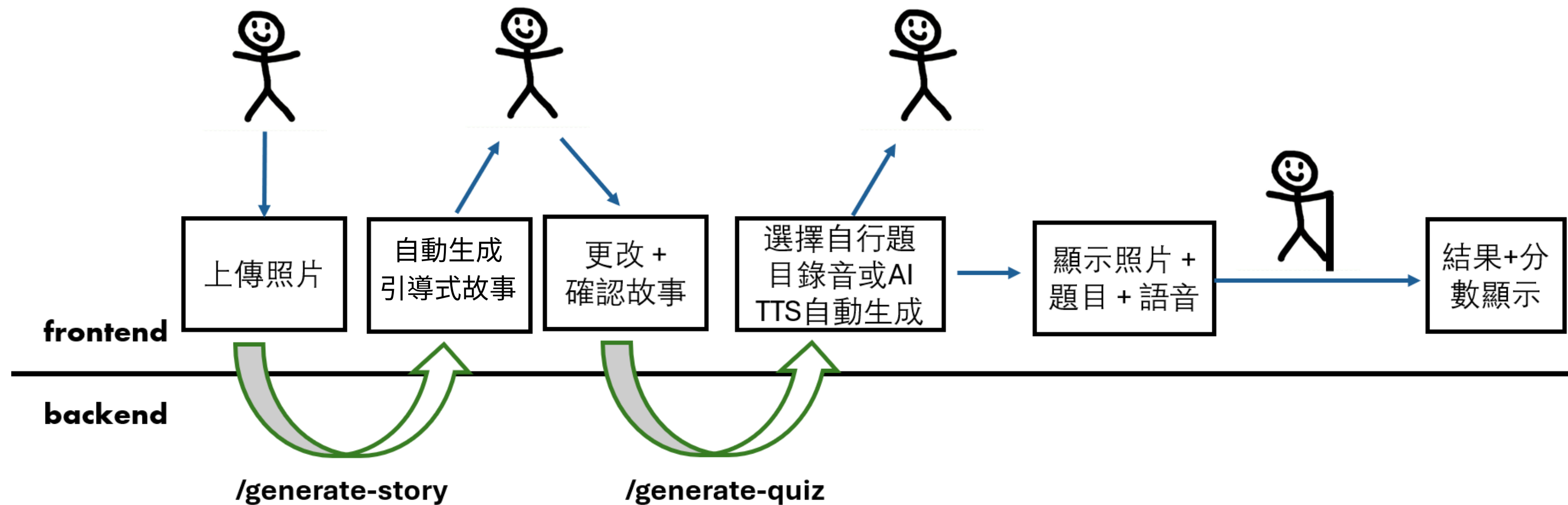


SELF-DEVELOPED

- **Frontend: React & TypeScript**
 - developed the elder-friendly UI/UX
 - handle picture upload & edit features
- **Backend: Node.js & Express**
 - central gateway for AI backend services
 - seamless integration of all third-party AI API
- **Data Handling**
 - client-side data handling via IndexedDB
 - implemented for privacy and performance

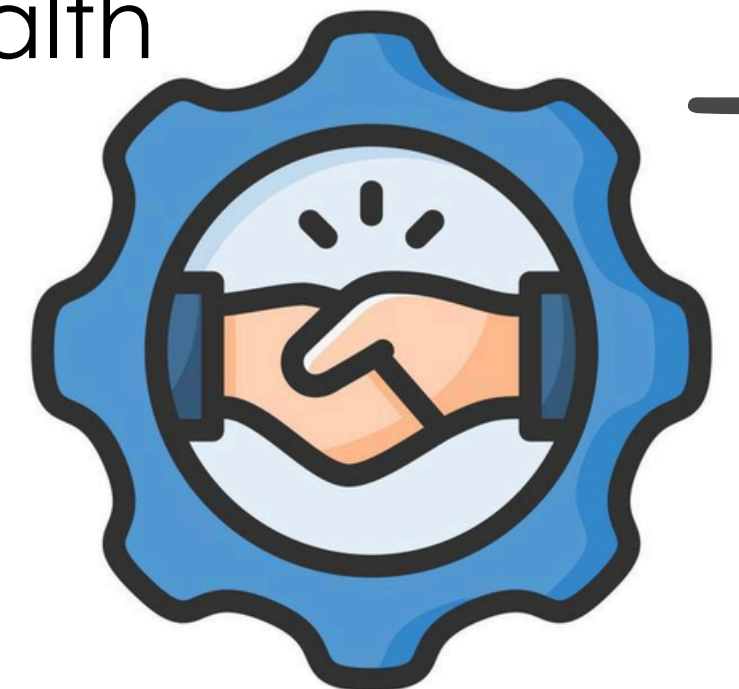


CORE FLOW



CONTRIBUTIONS

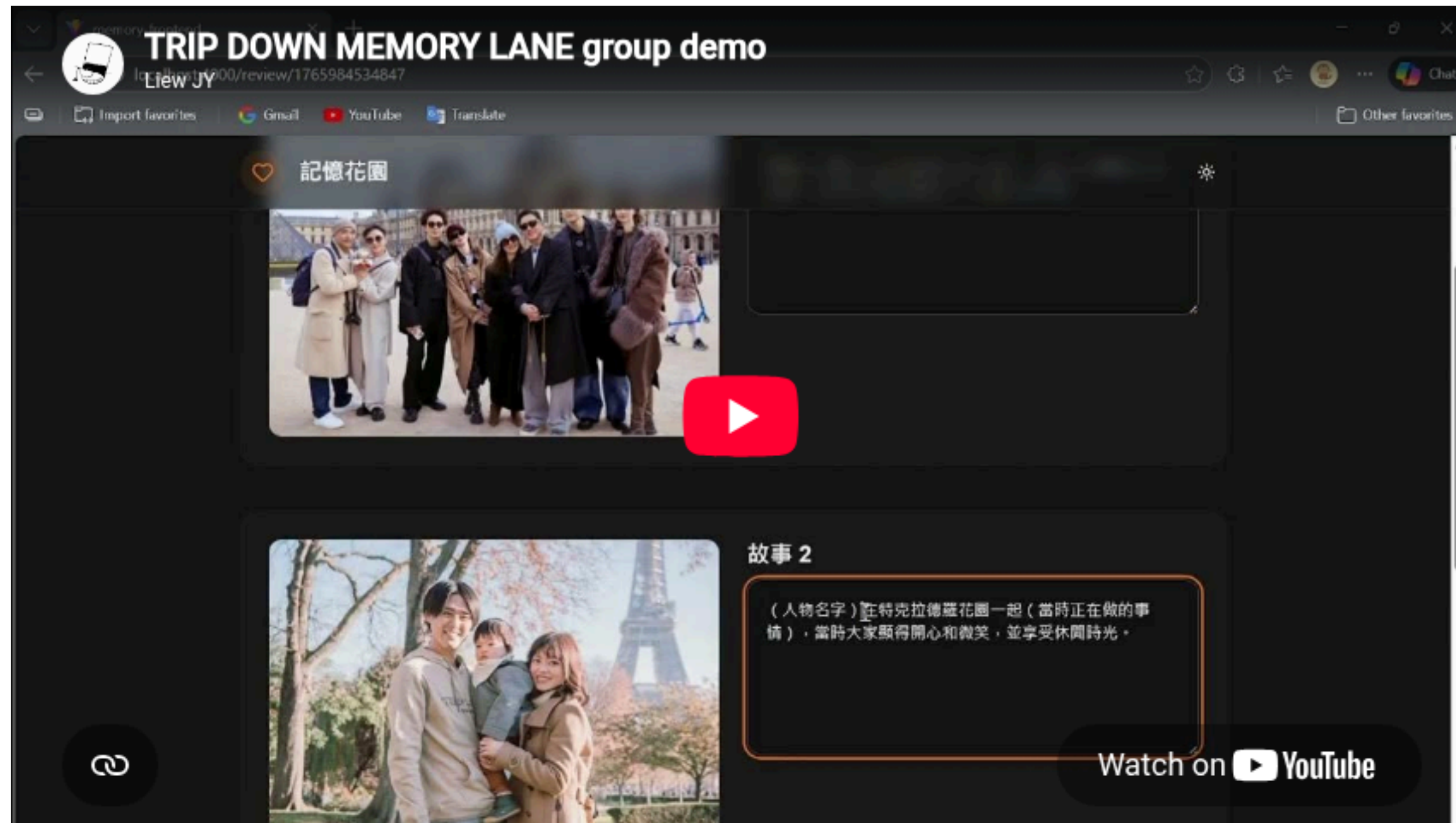
- **Novelty**
 - AI-generated, image-oriented quizzes for elders
- **System Integration**
 - seamless frontend, backend and indexedDB
- **Large Impact**
 - addressing a social need for elder cognitive health



TEAM WORK

MEMBER	TASK	
鄭俊良	Frontend: reviewStory page, quiz page	33.33%
陳達翔	Frontend: home page, addMemory page, IndexedDB	33.33%
劉靖媛	Backend: AI API connections & AI prompt	33.33%

DEMO



Frontend: <https://github.com/TripDownMemoryLane/memoryLaneFrontend.git>
Backend: <https://github.com/TripDownMemoryLane/quiz-api.git>